

SOME RESULTS ON RESTRICTED BAYES LEAST SQUARES PREDICTORS FOR FINITE POPULATIONS

Josemar Rodrigues : IME-USP, Rua do Matão, 1010 São Paulo, Brazil.

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Summary : The purpose of this paper is to introduce and examine the concept of *Restricted Bayes Least Squares* predictors for finite populations and to present some results based on distribution-free models. Some recent results in the literature are shown to be a natural consequence of this approach, and an example is considered in detail. Emphasis is given on the specification or not of the first and second moments of the distributions involved.

1. Introduction

The main purpose of this paper is to present a different distribution-free Bayesian approach for finite populations and to compare it with other recent Bayesian Least Squares approaches (Smouse, 1984; O'Hagan, 1984). The derivation of the Best Restricted Bayes Least Squares predictor requires only the first and second moments of the distributions involved.

The Restricted Bayes Least Squares predictors are studied under specific models and the results are outlined in table 1.

2. Background and notation

Consider a finite population of N identifiable units. Associated with the i -th unit is a fixed known p -dimensional column vector x_i and an unknown characteristic of interest y_i . We assume that $y' = (y_1, \dots, y_N)$ is a realization of a random vector Y which is related to the matrix $X = (x_1, \dots, x_N)'$ through the conditional superpopulation model,

$$(1) \quad \begin{aligned} (i) \quad & E[Y/\beta] = X\beta, \\ (ii) \quad & V[Y/\beta] = V, \end{aligned}$$

where β is a p -dimensional vector of unknown constants, and V is a known matrix. Suppose that the investigator's prior knowledge about β is

$$(2) \quad \begin{aligned} (iii) \quad & E[\beta/\mu] = \mu, \\ (iv) \quad & V[\beta/\mu] = \Sigma, \end{aligned}$$

where μ and Σ are known. The model defined by (i), (ii), (iii) and (iv) will be denoted by $M[V, \Sigma]$. The objective is to predict from a sample of size n and the model $M[V, \Sigma]$ the value of the function $\theta = \ell'Y$ for a specified $N \times 1$ vector $\ell' = (\ell_1, \dots, \ell_N)$.

Introducing the notation "s" for the sample and "r" for the remainder of the population, and listing the sampled units first, we may partition, Y, X, V, Ω and V^* in the following obvious fashion:

$$(3) \quad Y = \begin{bmatrix} Y_s \\ Y_r \end{bmatrix}, \quad X = \begin{bmatrix} X_s \\ X_r \end{bmatrix}, \quad V = \begin{bmatrix} V_s & V_{sr} \\ V_{rs} & V_r \end{bmatrix},$$

$$\Omega = \begin{bmatrix} \Omega_s & \Omega_{sr} \\ \Omega_{rs} & \Omega_r \end{bmatrix} = \begin{bmatrix} X_s \Sigma X_s' & X_s \Sigma X_r' \\ Y_r \Sigma X_s' & X_r \Sigma X_r' \end{bmatrix}, \text{ and}$$

$$\mathbf{V}^* = \begin{bmatrix} \mathbf{V}_s^* & \mathbf{V}_{sr}^* \\ \mathbf{V}_{rs}^* & \mathbf{V}_r^* \end{bmatrix} = \begin{bmatrix} \mathbf{V}_s + \boldsymbol{\Omega}_s & \mathbf{V}_{sr} + \boldsymbol{\Omega}_{sr} \\ \mathbf{V}_{rs} + \boldsymbol{\Omega}_{rs} & \mathbf{V}_r + \boldsymbol{\Omega}_r \end{bmatrix} = \mathbf{V} + \boldsymbol{\Omega}.$$

We denote $\mathbf{K}$ by the matrix $[\mathbf{V}_s, \mathbf{V}_{sr}]$ and $\mathbf{K}^* = [\mathbf{V}_s^*, \mathbf{V}_{sr}^*]$. Let $\mathbf{T}'$ be any n -dimensional vector,

$$(4) \quad L = \{\hat{\theta}: \hat{\theta} = \mathbf{a} + \mathbf{T}'\mathbf{Y}_s\},$$

the class of linear predictors of θ , and

$$(5) \quad U = \{\hat{\theta}: \hat{\theta} \in L \text{ and } E[\hat{\theta} - \theta] = 0\},$$

the class of Restricted Bayes Least Squares predictors (RBLS), where $E(\cdot)$ denotes expectation under the model $M[\mathbf{V}, \Sigma]$.

DEFINITION 1. Under the model $M[\mathbf{V}, \Sigma]$, $\hat{\theta}^*$ is said to be *Best Restricted Bayes Least Squares predictor of θ* (BRBLS) if

$$(6) \quad \begin{aligned} (i) \quad & \hat{\theta}^* = \mathbf{a}^* + \mathbf{T}^{*\prime}\mathbf{Y}_s \in U. \\ (ii) \quad & E[\hat{\theta}^* - \theta]^2 \leq E[\hat{\theta} - \theta]^2, \quad \hat{\theta} \in U. \end{aligned}$$

The proofs of the following lemmas can be obtained by standard computations and hence omitted. Definition 1 is suggested by lemma 2.

LEMMA 1. Under the model $M[\mathbf{V}, \Sigma]$, we have that

$$(7) \quad \hat{\theta} \in U \text{ iff } E[\mathbf{a} + \mathbf{F}'\mathbf{Y}_s - \mathbf{P}'\beta] = 0, \text{ where} \\ \mathbf{F}' = \mathbf{T}' - \ell'\mathbf{K}'\mathbf{V}_s^{-1} \text{ and } \mathbf{P}' = \ell'(\mathbf{X} - \mathbf{K}'\mathbf{V}_s^{-1}\mathbf{X}_s).$$

LEMMA 2. Under the model $M[\mathbf{V}, \Sigma]$, for all $\hat{\theta} = \mathbf{a} + \mathbf{T}'\mathbf{Y}$, we have

$$(8) \quad E[\hat{\theta} - \theta]^2 = \mathbf{V}[\mathbf{a} + \mathbf{F}'\mathbf{Y}_s - \mathbf{P}'\beta] + \ell'\mathbf{V}\ell - \ell'\mathbf{K}'\mathbf{V}_s^{-1}\mathbf{K}\ell + [\mathbf{a} + (\mathbf{T}'\mathbf{X}_s - \ell'\mathbf{X})\mu]^2,$$

where $\mathbf{F}'$ and $\mathbf{P}'$ are defined in lemma 1.

Lemma 2 is a Bayesian version of a result given by Tam (1986) and

provides the basic result for finding the BRBLS predictor of θ . The procedure to obtain the BRBLS is based heavily on the work of Rao (1971, p. 234) and is developed in section 2. When μ is unknown and fixed, the BRBLS predictor follows readily and this particular case is not considered in recent papers about Bayesian Linear Least Squares theory.

3. Restricted Bayes Least Squares Predictors

It is clear from lemma 1 and lemma 2 that the problem of finding $\hat{\theta}^*$ is equivalent to determining $a + F'Y_s$ such that

$$(9) \quad E[a + F'Y_s - P'\beta] = 0 \text{ and } V[a + F'Y_s - P'\beta] \text{ is a minimum.}$$

This problem can be solved in a straightforward manner by using Rao's results [1971, p. 234] in the following cases:

(i) μ known: The optimum choice of F and a are

$$(10) \quad \begin{aligned} F^* &= (V_s + X_s \Sigma X_s')^{-1} X_s \Sigma P \\ &= V_s^{-1} X_s [\Sigma^{-1} + X_s' V_s^{-1} X_s]^{-1} P, \\ a^* &= \mu' P - \mu' X_s' F^* \quad \text{and} \\ V[a^* - F^{*'} Y_s - P' \beta] &= P' (\Sigma^{-1} + X_s' V_s^{-1} X_s)^{-1} P. \end{aligned}$$

Formulas (10) provide the basic results for finding the BRBLS, that is,

$$(11) \quad \begin{aligned} \hat{\theta}^* &= a^* + T^{*'} Y_s \\ &= \ell_s' Y_s + \ell_r' [X_r \bar{\beta} + V_{rs} V_s^{-1} (Y_s - X_s \bar{\beta})], \end{aligned}$$

where $\bar{\beta} = C \hat{\beta}_V + (I - C) \mu$, $\hat{\beta}_V = (X_s' V_s^{-1} X_s)^{-1} X_s' V_s^{-1} Y_s$,

$$C = [\Sigma^{-1} + X_s' V_s^{-1} X_s]^{-1} (X_s' V_s^{-1} X_s) \quad \text{and}$$

$$E[\hat{\theta}^* - \theta]^2 = P' (\Sigma^{-1} + X_s' V_s^{-1} X_s)^{-1} P + \ell' V \ell - \ell' K' V_s^{-1} K \ell$$

(ii) μ unknown and fixed: The optimum choice of F and a are:

$$(12) \quad F^* = V_s^{-1} X_s (X_s' V_s^{-1} X_s)^{-1} P, \quad a^* = 0 \quad \text{and} \\ V[F^* Y_s - P' \beta] = P' (X_s' V_s^{-1} X_s)^{-1} P.$$

Consequently, we have that the BRBLS is given by

$$(13) \quad \hat{\theta}^* = \ell_s' Y_s + \ell_r' [X_r \hat{\beta}_V + V_{rs} V_s^{-1} (Y_s - X_s \hat{\beta}_V)] \quad \text{and} \\ E[\hat{\theta}^* - \theta]^2 = P' (X_s' V_s^{-1} X_s)^{-1} P + \ell' V \ell - \ell' K V_s^{-1} K \ell.$$

Observe that $\hat{\theta}^*$ given by (13) was obtained by Royall & Pfeiffermann (1982) using the multivariate normal distribution and the diffuse prior distribution for β . The result (11) was also considered by Pereira *et al.* (1986) and Malec *et al.* (1985) using normality assumptions.

In order to relate $\hat{\theta}^*$ to other classes of Bayesian Least Squares predictors, let us consider the following particular cases of the model $M[V, \Sigma]$:

$$(14) \quad M_V = M[V, O], \quad M_\Sigma = M[O, \Sigma], \\ \bar{M}_V = M[V, \Sigma^{-1} = O], \quad \bar{M}_\Sigma = M[V^{-1} = O, \Sigma].$$

Consider the following models examined by Smouse 1984:

$$(15) \quad H[V, \Sigma]: \begin{cases} E[Y/\mu] = X\mu \\ V[Y/\mu] = V^* \end{cases} \\ H_V = H[V, O] \quad \text{and} \quad H_\Sigma = H[O, \Sigma].$$

Note that $H_V = M_V$, $H_\Sigma = M_\Sigma$. If the distribution of Y given β is independent of μ then $M[V, \Sigma]$ implies $H[V, \Sigma]$. The models $\bar{M}_\Sigma$ and $\bar{M}_V$ are noninformative w.r.t. β and μ , respectively. The model M_Σ with $X = I$ (identity matrix) is discussed by O'Hagan (1984) and the models $H[V, \Sigma]$ and H_V are discussed by Smouse (1984) from the coordinate-free point of view. In these papers the parameter μ is considered known. Consider two predictors. First, under $H[V, \Sigma]$, it is

Smouse's predictor (1984), namely

$$(16) \quad \hat{Y}_S = \ell'_s Y_s + \ell'_r [X_r \mu + V_{rs}^* V_s^{*-1} (Y_s - X_s \mu)],$$

which after some algebraic simplifications is equal to (11).

Next, under M_Σ and $X = I(p > n)$, it is O'Hagan's predictor:

$$(17) \quad \hat{Y}_O = \ell'_s y_s + \ell'_r [\mu + \Omega_{rs} \Omega_s^{-1} (Y_s - \mu)].$$

Observe that $\hat{Y}_S$ and $\hat{Y}_O$ are BRBLS under their respective models.

Table 1, when μ is known, outlines the results obtained from the Restricted Bayesian Least Squares point of view.

TABLE 1. RESTRICTED BAYESIAN
LEAST SQUARES PREDICTORS. μ : KNOWN.

Models	BRBLS predictors
$M[V, \Sigma]$	$\ell'_s Y_s + \ell'_r [X_r \hat{\beta} + V_{rs} V_s^{-1} (Y_s - X_s \hat{\beta})]$
MV	$\ell'_s Y_s + \ell'_s [X_r \mu + V_{rs} V_s^{-1} (Y_s - X_s \mu)]$
M_Σ	$\ell'_s Y_s + \ell'_r [\mu + \Omega_{rs} \Omega_s^{-1} (Y_s - \mu)] \quad (X = I)$
$\bar{M}_V$	$\ell'_s Y_s + \ell'_r [X_r \hat{\beta}_V + V_{rs} V_s^{-1} (Y_s - X_s \hat{\beta}_V)]$
M_Σ	$\ell'_s Y_s + \ell'_r X_r \mu$
$H[V, \Sigma]$	$\ell'_s Y_s + \ell'_r [X_r \hat{\beta} + V_{rs} V_s^{-1} (Y_s - X_s \hat{\beta})]$

4. An example

Consider the model $M[V, \Sigma]$ with

$$\begin{aligned}
 (18) \quad \mathbf{X}' &= (1, \dots, 1) \\
 \mathbf{V} &= \sigma_0^2 \mathbf{I} \\
 \Sigma &= \sigma_0^2 \mathbf{I} \\
 \mathbf{t}' &= \left(\frac{1}{N}, \dots, \frac{1}{N} \right).
 \end{aligned}$$

It is not difficult to see from (11), that

$$(19) \quad \hat{\theta}^* = \left(\frac{\frac{\sigma_0^2}{N} + \sigma_0^2}{\frac{\sigma_0^2}{n} + \sigma_0^2} \right) \bar{Y}_s + \left[1 - \left(\frac{\frac{\sigma_0^2}{N} + \sigma_0^2}{\frac{\sigma_0^2}{n} + \sigma_0^2} \right) \right] \mu, \quad \bar{Y}_s = \frac{\sum Y_i}{n},$$

and

$$E[\hat{\theta}^* - \theta]^2 = \left(1 - \frac{n}{N} \right) \frac{\sigma_0^2 (N+n) \sigma_0^2 + \sigma_0^2}{N \left(\frac{\sigma_0^2}{n} + \sigma_0^2 \right)}.$$

Goldstein (1975), with the purpose of estimating the super parameter β , considered the Bayes Linear estimator,

$$(20) \quad \hat{\theta}_G = \frac{\frac{\sigma_0^2}{n} + \sigma_0^2}{\frac{\sigma_0^2}{n} + \sigma_0^2} \bar{Y}_s + \frac{\frac{\sigma_0^2}{n}}{\frac{\sigma_0^2}{n} + \sigma_0^2} \mu$$

$$\text{and} \quad E[\hat{\theta}_G - \beta]^2 = \left(\frac{n}{\sigma_0^2} + \frac{1}{\sigma_0^2} \right)^{-1}.$$

Next, we have interesting results w.r.t. $\hat{\theta}^*$:

$$(a) \quad \hat{\theta}^* = \frac{n}{N} \bar{Y}_s + \left(1 - \frac{n}{N} \right) \hat{\theta}_G,$$

$$(b) \quad \text{If } E[\beta / Y_s] = a \bar{Y}_s + b, \text{ then}$$

$$(1) \quad E[\theta / Y_s] = \hat{\theta}^*,$$

$$(2) \quad V[\theta / Y_s] \leq E[\hat{\theta}^* - \theta]^2 \iff \frac{(N-n-1)}{n} \sigma_0^2 \geq \sigma_0^2,$$

$$(c) \quad \lim_{N \rightarrow \infty} E[\hat{\theta}^* - \theta]^2 = E[\hat{\theta}_G - \beta]^2,$$

- (d) If either $\sigma_0^2 \rightarrow 0$ or $\sigma^2 \rightarrow \infty \rightarrow \hat{\theta}^* = \frac{n}{N} \bar{Y}_s + (1 - \frac{n}{N})\mu$ (BRBLS),
- (e) If either $\sigma_0^2 \rightarrow \infty$ or $\sigma^2 \rightarrow 0 \rightarrow \hat{\theta}^* = \bar{Y}_s$ (Royall & Pfefferman, 1982; O'Hagan, 1984).

The predictor β_V , which is a weighted average of the prior mean μ and the sample mean $\bar{Y}_s$, is the *l.s.* predictor of each unobserved Y_i based on Y_s . The result (e) was obtained by O'Hagan (1984) using unrestricted Bayes linear approach and by Ericson (1969a) under the assumption of normality. The result (b) (1) is similar to the result obtained by Goldstein (1975), theorem 1.

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References

- ERICSON, W.A. (1969a) Subjective Bayesian Models in sampling finite populations (with discussion). *J. Roy. Statist. Soc. B*—31, 195–233.
- GOLDSTEIN, M. (1975). A note on some Bayesian non-parametric estimates. *Ann. Statist.* 3, 736–740.
- MALEC, D. & SEDRANSK, J. (1985). Bayesian Inference for Finite Population Parameters in Multistage Cluster Sampling. *JASA*, vol. 80, no. 392, 897–903.
- O'HAGAN, A. (1984). Bayes Linear Estimators for Finite Populations. Report no. 58 – University of Warwick, England.
- PEREIRA, C.A.B.; BOLFARINE, H.; RODRIGUES, J. (1986). Robust Linear Prediction in Finite Populations: A Bayesian Perspective. *Sankhyā*, Series B, vol. 49 pt. 1, pp. 23–35.

- ROYALL, P.M. & PFEFFERMANN, D. (1982). Balanced samples and Robust Bayesian inference in population sampling. *Biometrika* 69, 401–409.
- RAO, C.R. (1971). *Linear Statistical Inference and its Applications*. New York. Wiley.
- SMOUSE, E.P. (1984). A note on Bayesian Least Squares inference for Finite Population Models. *JASA*, vol. 79, no. 396. 390–392.
- TAM, S.M. (1986). Characterization of best model-based predictors in survey sampling. *Biometrika*, 73, 1, 232–235.